

<page_start no = 1>

<sol_start id=1>

1. $\text{LCM}(36, 45) = 2^2 \times 3^2 \times 5 = 180 \text{ Sec} = 3 \text{ min}$

$$36 = 2^2 \times 3^2$$

$$45 = 3^2 \times 5$$

Ans. : $\boxed{(3) 3}$

<sol_end>

<sol_start id=2>

2. $\frac{2 \tan 30^\circ}{1 - \tan^2 30^\circ}$

$$\Rightarrow \frac{2 \left(\frac{1}{\sqrt{3}} \right)}{2 - \left(\frac{1}{\sqrt{3}} \right)^2}$$

$$= \frac{\frac{2}{\sqrt{3}}}{3 - 1}$$

$$= \frac{2 / \sqrt{3}}{2 / 3}$$

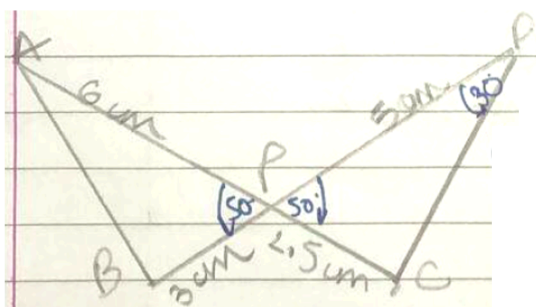
$$= \frac{\frac{1}{\sqrt{3}}}{\frac{1}{3}}$$

$$= \frac{3}{\sqrt{3}} = \sqrt{3}$$

Ans. : $\boxed{(3) \tan 60^\circ}$

<sol_end>

<sol_start id=3>



3.

$$\angle APB = \angle CPD = 50$$

(vertically opposite angles)

$$\frac{PA}{PD} = \frac{PB}{PC} = \frac{6}{5} = \frac{3}{2.5} = 1.2$$

By SAS similarity $\triangle APB \sim \triangle DPC$

$$\angle PAB = \angle PDC = 30^\circ$$

$$\angle PBA = 180 - (50 + 30)$$

$$= 180 - 80 = 100^\circ$$

$$\text{Ans. : } \boxed{(4)100^\circ}$$

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<sol_start id=4>

$$4. \quad \triangle ABC \sim \triangle QRP$$

$$\frac{\text{au}(ABC)}{\text{au}(PQR)} = \frac{9}{4} = \frac{BC^2}{PR^2}$$

$$\frac{9}{4} = \frac{(15)^2}{(PR)^2}$$

$$\frac{9}{4} = \frac{225}{PR^2}$$

$$PR^2 = 100$$

$$PR = 10 \text{ un}$$

$$\text{Ans. : } \boxed{(1)10\text{un}}$$

<sol_end>

<sol_start id=5>

$$5. \quad P(x) = x^2 + (a+1)x + b$$

$$P(2) = (2)^2 + (a+1)^2 + b$$

$$0 = 4 + 2a + 2 + b$$

$$0 = 6 + 2a + b$$

$$(-6) = 2a + b \dots\dots(1)$$

$$P(-3) = (-3)^2 + (a + 1)(-3) + b$$

$$0 = 9 - 3a - 3 + b$$

$$-6 = b - 3a \dots\dots(2)$$

Equating equation 1 and equation 2

$$2a + \cancel{b} = \cancel{b} - 3a$$

$$\sqrt{a} = 0$$

Ans. : $a = 0$

<sol_end>

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<page_start no = 3>

<sol_start id=6>

$$6. \quad \frac{14587}{2 \times 5^4}$$

$$\text{Denominator} = 2^m \times 5^n$$

$$2^1 \times 5^n = 2^m \times 5^n$$

$$m = 1$$

$$n = 4 \quad \& \quad m < n$$

the no. will terminate after 4 decimal places

Ans. : (4) four decimal places.

<sol_end>

<sol_start id=7>

$$7. \quad x = a$$

$$y = b$$

$$x - y = 2$$

$$a - b = 2 \dots\dots(1)$$

$$x + y = 4$$

$$a + b = 4 \dots\dots(2)$$

add equation 1 and equation 2

$$a - \cancel{b} + a + \cancel{b} = 2 + 4$$

$$2a = 6$$

$$a = 3$$

Put a in equation 1

$$3 - 2 = b$$

$$1 = b$$

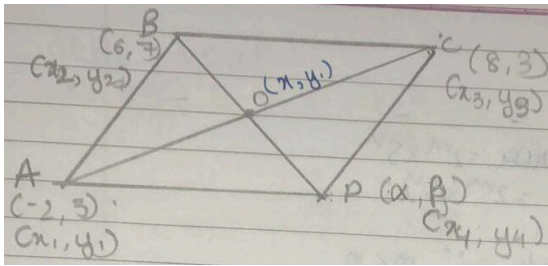
Ans. : (3) 3 and 1

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<page_start no = 4>

<sol_start id=8>



8.

Join AC and BD and intersect at O.

▯ diagonals bisect each other in a 11^{gm}

O is mid point of both the diagonals

$$x = \frac{x_2 + x_4}{2} = \frac{x_1 + x_3}{2}$$

$$\left[\frac{6 + \cancel{\beta}}{2} = \right]$$

$$\frac{6 + \cancel{\beta}}{2} = \frac{(-2) + 8}{2}$$

$$6 + \cancel{\beta} = 6$$

$$\cancel{\beta} = 0$$

$$y = \frac{y_2 + y_4}{2} = \frac{y_3 + y_1}{2}$$

$$\frac{7 + \beta}{2} = \frac{3 + 3}{2}$$

$$7 + \beta = 6$$

$$\beta = (-1)$$

$$\text{Ans. : } \boxed{(2)(0, -1)}$$

<sol_end>

<page_end>

<page_start no = 5>

<sol_start id=9>

9. $9x + 3y + 12 = 0$

$$18x + 6y + 26 = 0$$

$$\left[\frac{\cancel{a_1}}{\cancel{b_2}} = \frac{\cancel{b_1}}{\cancel{b_2}} = \frac{\cancel{c_1}}{\cancel{c}} \right] \frac{1}{2} \frac{\cancel{9}}{18} = \frac{\cancel{3}}{\cancel{6}} \frac{1}{2} \neq \frac{12}{26}$$

$$\frac{1}{2} = \frac{1}{2} \neq \frac{6}{13}$$

$$\text{Ans. : } \boxed{(4)\text{nosolution}}$$

<sol_end>

<sol_start id=10>

10. $2x - 3y + 10 = 0$

$$3x + ky + 15 = 0$$

As coincident lines

$$\frac{2}{3} = \frac{(-3)}{k} = \frac{10}{15}$$

$$\frac{2}{3} = \frac{(-3)}{k}$$

$$k = \left(\frac{-9}{2} \right)$$

Ans. :

$$(1) \ k = \left(\frac{-9}{2} \right)$$

<sol_end>

<sol_start id=11>

11. $10x - 5y + 15 = 0$

$$6x + ky + 19 = 0$$

As intersecting lines

$$\frac{5}{3} \neq \frac{(-5)}{k}$$

$$k \neq -3$$

Ans. :

$$(1) \ k \neq -3$$

<sol_end>

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<page_start no = 6>

<sol_start id=12>

12. $\text{HCF}(6, 6, 32) = 2^3 = 8$

$$616 = 2^3 \times 7 \times 11$$

$$32 = 2^5$$

$\therefore$ 8 rows

Ans. :

$$(1) \ 8$$

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<sol_start id=13>

13. $(1) \text{ similar}$

<sol_end>

<sol_start id=14>

14. $a = x^3 y^4 z^2$

$$b = x^2 y z^4$$

$$\text{LCM}(a, b) = x^3 y^4 z^4$$

Ans.: $(1) x^3 y^4 z^4$

<sol_end>

<sol_start id=15>

15. $\alpha = (-3)$

$\beta = (4)$

$\alpha + \beta = -3 + 4 = 1$

$\alpha \beta = (-3)4 = (-12)$

$k[x^2 - x(\alpha + \beta) + \alpha \beta]$

$k[x^2 - x - 12]$

$k\left[\frac{x^2}{2} - \frac{x}{2} - 6\right]$

Ans.: $(3) \frac{x^2}{2} - \frac{x}{2} - 6$

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<sol_start id=16>

16. $(3) \text{intersecting or coincident}$

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<page_start no = 7>

<sol_start id=17>

17. $(1) \text{parallel}$

<sol_end>

<sol_start id=18>

18. Let $\Delta ABC \sim \Delta PQR$

$$\frac{\text{ar}(ABC)}{\text{ar}(PQR)} = \frac{AB^2}{PQ^2} = \frac{(4)^2}{(9)^2} = \frac{16}{81}$$

Ans. $(1) 16:81$

<sol_end>

<sol_start id=19>

19. A: $x^2 - 4x - 5$

$x^2 - 5x + x - 5 = 0$

$x(x - 5) + (x - 5) = 0$

$(x + 1)(x - 5) = 0$

$x = 5; (-1)$

$$\begin{aligned}
 R: \quad \alpha &= 2 + \sqrt{3} \\
 \beta &= 2 - \sqrt{3} \\
 \alpha + \beta &= 4 \\
 \alpha\beta &= (4 - 3) = 1 \\
 P(x) &= x^2 - 4x + 1
 \end{aligned}$$

Ans. (3) Assertion (A) is true but Reason R is false

<sol_end>

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<page_start no = 8>

<sol_start id=20>

$$\begin{aligned}
 20. \quad 3(k-1)x + 4y - 24 &= 0 \\
 15x + 20y - 8(k+13) &= 0
 \end{aligned}$$

$$\frac{\cancel{3}(k-1)}{\cancel{15}} = \frac{\cancel{4}}{\cancel{20}} \frac{1}{5}$$

$$k - 1 = 1$$

$$k = 2$$

Ans. (1) Both A and R are true and R is correct explanation of A

<sol_end>

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<page_start no = 9>

<sol_start id=21>

21. To prove :

$$6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5 \text{ is a composite no.}$$

$$725 = 5 \times 5 \times 29 \times 1$$

It has more than two factors the number is composite number

Hence proved

<sol_end>

<sol_start id=22>

$$22. \quad 4x + 6y = 7 \Rightarrow 4x + 6y - 7 = 0$$

$$kx + 18y = 21 \Rightarrow kx + 18y - 21 = 0$$

$$a_1 = 4 \quad b_1 = 6 \quad c_1 = (-7)$$

$$a_2 = k \quad b_2 = 18 \quad c_2 = (-21)$$

As a system of linear equation of infinite solutions.

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$\frac{4}{k} = \frac{6}{18}$$

$$\frac{4}{k} = \frac{1}{3}$$

$$k = 12$$

<sol_end>

<page_end>

<page_start no = 10>

<sol_start id=23>

23. Let distance between origin and $(-6, 8)$ be AB

$$x_1 = 0 \quad y_1 = 0$$

$$x_2 = (-6) \quad y_2 = 8$$

Applying distance formula

$$\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} = AB$$

$$\sqrt{(0 - (-6))^2 + (0 - 8)^2} = AB$$

$$\sqrt{36 + 64} = AB$$

$$\sqrt{100} = AB$$

$$AB = 10 \text{ u}$$

Ans. $\therefore$ distance between origin and $(-6, 8)$ is 10u

<sol_end>

<sol_start id=24>

$$24. \sin 3A = \cos(A-10)$$

$$\sin A = \cos(90-A)$$

$$\text{so } \sin 3A = \cos(90-3A)$$

$$\cos(90-3A) = \cos(A-10)$$

on comparing angles.

$$90-3A = A-10$$

$$100 = 4A$$

$$\text{Ans. } 25 = A$$

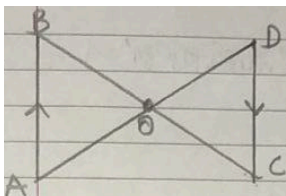
$\therefore$ A is equal to 25.

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<page_start no = 11>

<sol_start id=25>



25.

given $AB \parallel CD$.

to prove $\triangle AOB \cong \triangle DOC$

proof $AB \parallel CD$

$$\angle ABO = \angle DCO \text{ [alternate interior angles]}$$

$$\angle BAO = \angle CDO \text{ [alternate interior angles]}$$

By AA similarity criterion

$$\angle AOB \cong \angle DOC$$

Hence proved.

<sol_end>

<page_end>

<page_start no = 12>

<sol_start id=26>

26. To prove :

$$\sin(50 + \theta) - \cos(40 - \theta) + \tan 1^\circ \cdot \tan 10^\circ \cdot \tan 80^\circ \cdot \tan 89^\circ = 1$$

$$\text{proof } \sin(50 + \theta) = \cos(90 - (50 + \theta))$$

$$= \cos(40 - \theta)$$

$$\tan 1^\circ = \cos(90 - 1^\circ)$$

$$= \cot 89^\circ$$

$$\tan 10^\circ = \cot(90 - 10^\circ)$$

$$= \cot 80^\circ$$

$$\sin(50 + \theta) - \cos(40 - \theta) + \tan 1^\circ \cdot \tan 10^\circ \cdot \tan 80^\circ \cdot \tan 89^\circ$$

$$\cos(40 - \theta) - \cos(40 - \theta) + \cot 89^\circ \cdot \tan 89^\circ \cdot \cot 80^\circ \cdot \tan 80^\circ$$

$$0 + (1)(1) \left[\tan \theta \times \cot \theta = \tan \theta \times \frac{1}{\tan \theta} = 1 \right]$$

$$0 + 1$$

$$1.$$

Hence proved.

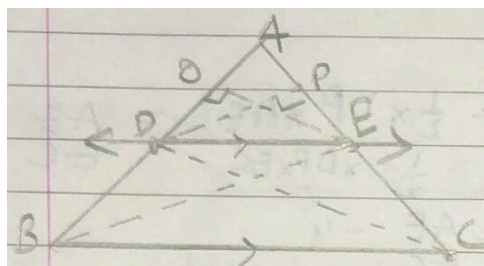
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<page_start no = 13>

<sol_start id=27>

27. Basic proportionality theorem states that. If a line is drawn parallel to one side of triangle intersecting the other two sides in distinct points then other two sides are divided in same ratio.



given $ED \parallel BC$

$$\text{to prove } \frac{AD}{BD} = \frac{AE}{EC}$$

construct (1) Join BE and CD.

(2) then draw a perpendicular EO such that $AD \perp EO$.

(3) draw a perpendicular $DP \perp AE$.

$$\text{area of } \triangle ADE = \frac{1}{2} \times AD \times EO \quad \dots(1)$$

$$\text{area of } \triangle BDE = \frac{1}{2} \times BD \times EO \quad \dots(2)$$

dividing equation 1 by equation 2

$$\frac{\text{area}(\triangle ADE)}{\text{area}(\triangle BDE)} = \frac{\frac{1}{2}(AD \times EO)}{\frac{1}{2}(BD \times EO)} = \frac{AD}{BD} \quad \dots(3)$$

$$\frac{\text{area}(\triangle ADE)}{\text{area}(\triangle CDE)} = \frac{\frac{1}{2} \times DP \times AE}{\frac{1}{2} \times DP \times EC} = \frac{AE}{EC}$$

similarly :

$$\frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle CDE)} = \frac{AE}{EC} \quad \dots(4)$$

$$\text{area } [\triangle BDE] = \text{area } [\triangle CDE] \quad \dots(5)$$

[∵ they lie on same base DE and between 11^e lines]
from equation 4 and 5.

$$\frac{\text{area}(\triangle ADE)}{\text{area}(\triangle BDE)} = \frac{AE}{EC} \quad \dots(6)$$

from equation 3 and equation 6.

$$\frac{AD}{DB} = \frac{AE}{EC}$$

Hence proved.

<sol_end>

<page_end>

<page_start no = 14>

<sol_start id=28>

28. Let

fixed charge be 'x' rupees

extra charge be 'y' rupees

(per day)

Case-I : Latika paid ₹22

$$x + 4y = 22 \quad \dots(1)$$

Case-II: Anand paid ₹16

$$x + 2y = 16 \quad \dots(2)$$

subtract equation 2 from equation 1.

$$x + 4y - [x + 2y] = 22 - 16.$$

$$x + 4y - x - 2y = 6$$

$$2y = 6$$

$$y = ₹3$$

substitute y = 3 in equation 1

$$x + 4(3) = 22$$

$$x = 22 - 12$$

$$x = 10$$

$\therefore$ fixed charges is ₹10/- and extra charge per day is ₹3/-

<sol_end>

<page_end>

<page_start no = 15>

<sol_start id=29>

29. Area of quadrilateral ABCD = area of $\triangle ABC$ + area of $\triangle ADC$

$$= \frac{35}{2} + \frac{109}{2}$$

$$= \frac{35+109}{2}$$

$$= \frac{144}{2}$$

$$= 72 \text{ u}^2$$

$\therefore$ area of quadrilateral is $\boxed{72\text{u}^2}$.

<sol_end>

<sol_start id=30>

30. HCF (1353, 861) =

$$1353 = 861 \times 1 + 492$$

$$861 = 492 \times 1 + 369$$

$$492 = 369 \times 1 + 123$$

$$369 = 123 \times 3 + 0$$

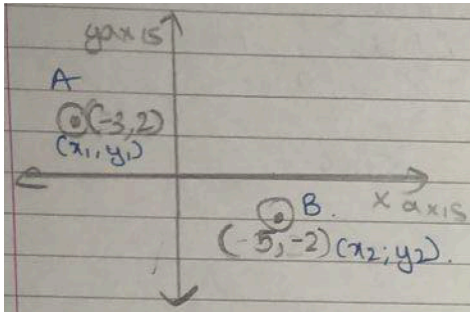
$\therefore$ HCF of 1352 and 861 is $\boxed{123}$

<sol_end>

<page_end>

<page_start no = 16>

<sol_start id=31>



31.

$$A = (-3, 2)$$

$$B = (5, -2)$$

$$D = (x, 0)$$

$$x_1 = (-3) \quad y_1 = 2$$

$$x_2 = 5 \quad y_2 = (-2)$$

$$AD = BD$$

Applying distance formula

$$\sqrt{(x_1 - x_3)^2 + (y_1 - y_3)^2} = \sqrt{(x_2 - x_3)^2 + (y_2 - y_3)^2}$$

Squaring on both the sides

$$(x_1 - x_3)^2 + (y_1 - y_3)^2 = (x_2 - x_3)^2 + (y_2 - y_3)^2$$

$$(-3 - x)^2 + (2 - (0))^2 = (5 - x)^2 + (-2)^2$$

$$\left[\begin{array}{l} (a - b)^2 = a^2 + b^2 - 2ab \\ (a + b)^2 = a^2 + b^2 + 2ab \end{array} \right]$$

$$9 + x^2 - 6x + 4 = 25 - 10x + x^2 + 4$$

$$x^2 - 6x = 16$$

$$x^2 - 6x - 16 = 0$$

$$x^2 - 10x + 4x - 16 = 0$$

$$x(x - 10) + 4(x - 10) = 0$$

$$(x - 10)(x + 4) = 0$$

$$x - 10 = 0 \quad | \quad x + 4 = 0$$

$$x = 10 \quad | \quad x = (-4)$$

$\therefore$ points on x axis are $\boxed{10}$ and $\boxed{-4}$.

<sol_end>

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<page_start no = 17>

<sol_start id=32>

$$32. \quad \frac{\tan \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 - \tan \theta}$$

$$\frac{\frac{\sin \theta}{\cos \theta}}{1 - \frac{\cos \theta}{\sin \theta}} + \frac{\frac{\cos \theta}{\sin \theta}}{1 - \frac{\sin \theta}{\cos \theta}} \left[\tan \theta = \frac{\sin \theta}{\cos \theta} \cot \theta = \frac{\cos \theta}{\sin \theta} \right]$$

$$\frac{\sin^2 \theta}{\cos \theta (\sin \theta - \cos \theta)} + \frac{\cos^2 \theta}{\sin \theta (\cos \theta - \sin \theta)}$$

$$\frac{\sin^2 \theta}{\cos \theta (\sin \theta - \cos \theta)} - \frac{\cos^2 \theta}{\sin \theta (\sin \theta - \cos \theta)}$$

$$\frac{\sin^3 \theta - \cos^3 \theta}{\sin \theta \cos \theta (\sin \theta - \cos \theta)}$$

$$\frac{(\sin \theta - \cos \theta)(\sin^2 \theta + \cos^2 \theta + \sin \theta \cos \theta)}{(\sin \theta \cos \theta)(\sin \theta - \cos \theta)} \left[a^3 - b^3(a-b)(a^2 + b^2 + ab) \right]$$

$$1 + \frac{\sin \theta \cos \theta}{\sin \theta \cos \theta} \quad \left[\sin^2 \theta + \cos^2 \theta = 1 \right]$$

$$1 + \sec \theta \operatorname{cosec} \theta$$

Hence proved

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<sol_start id=33>

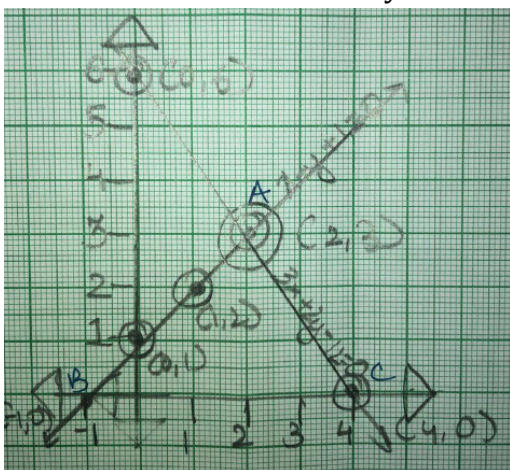
33. $x - y + 1 = 0$

If $x = 2$ if $x = 0$ if $x = 1$
 $2 - y + 1 = 0$ $0 - y + 1 = 0$ $1 - y + 1 = 0$
 $3 = y$ $1 = y$ $2 = y$

$x - y + 1$			
	-0		
x	0	1	2
y	1	2	3

$3x + 2y - 12$			
	-0		
x	0	2	4
y	6	3	0

if $x = 2$ $3x + 2y - 12 = 0$ if $x = 4$
 $12 + 2y - 12 = 0$
 $2y = 0$
 $y = 0$



Virtu us of $\triangle ABC$

A - (2, 3)

B - (-1, 0)

C - (4, 0)

$$\begin{array}{ll} x_1 = 2 & y_1 = 3 \\ x_2 = (-1) & y_2 = 0 \\ x_3 = 4 & y_3 = 0 \end{array}$$

$$\begin{aligned} \text{area of } \Delta ABC &= \frac{1}{2} \left[x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) \right] \\ &= \frac{1}{2} \left[2[(0-0)] + [-1(0-3)] + [(4-0)] \right] \\ &= \frac{1}{2} [0 + 3 + 4] \\ &= \frac{7}{2} u^2 \\ &= 3.5u^2 \end{aligned}$$

$$\therefore \text{ area of } \Delta = 3.5u^2$$

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<sol_start id=34>

$$\begin{aligned} 34. \quad x &= \sqrt{2} & x &= -\sqrt{2} \\ x - \sqrt{2} &= 0 & x + \sqrt{2} &= 0 \\ (x - \sqrt{2})(x + \sqrt{2}) &= x^2 - 2 \quad [a^2 - b^2 = (a-b)(a+b)] \end{aligned}$$

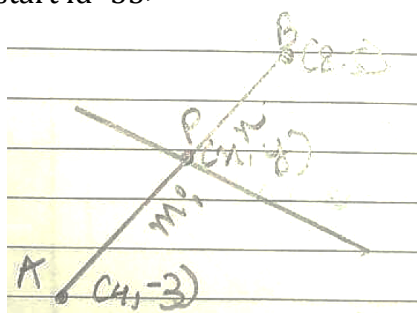
$$\begin{array}{r} 2x \\ x \overline{) 2x^3 - 4x - x^2} \\ \underline{2x^3} \\ -4x - x^2 \\ \underline{-4x} \\ 0 - x^2 \\ + 2x^2 \\ \underline{+2x^2} \\ 0 \end{array} \quad \text{so } 2x - 1 = 0 \quad x = \frac{1}{2}$$

all zeroes of polynomial $2x^3 - 4x - x^2 + 2$ are $\sqrt{2}$, $-\sqrt{2}$ and $\frac{1}{2}$.

<sol_end>

<sol_start id=35>

35.



$$m : n = 3 : 1$$

$$m = 3x$$

$$n = x$$

$$x_1 = 4$$

$$x_2 = 8$$

$$y_1 = (-3)$$

$$x = \frac{mx_2 + nx_1}{m + x}$$

$$= \frac{3x(8)}{3x + x} + x(4)$$

$$= \frac{3x(6+1)}{4x} \Rightarrow x = 7$$

$$y_2 = 5$$

$$y = \frac{my_2 + ny_1}{m + x}$$

$$= \frac{3x(5)}{3x + x} + x(-3)$$

$$= \frac{3x(5-1)}{4x} \Rightarrow 3 = y$$

Coordinates of point P is 7,3

<sol_end>

<sol_start id=36>

36. (i) B (0, 12)

Ans. (2) (0, 12)

(ii) P (3, 5)

Ans. (1) (3, 5)

(iii) Q (6, 9)

P (3, 5)

$$x_1 = 3 \quad y_1 = 5$$

$$x_2 = 6 \quad y_2 = 9$$

Applying distance formula

$$\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} = PQ$$

$$\sqrt{(3-6)^2 + (5-9)^2} = PQ$$

$$\sqrt{(3)^2 + (-4)^2} = PQ$$

$$\sqrt{9+16} = PQ$$

$$\sqrt{25} = PQ$$

$$\boxed{PQ = 5m}$$

(3) 5 m

<sol_end>

<page_end>

<page_start no = 19>

<sol_start id=37>

37. (i) (3) Parabola

(ii) (3) 0

(iii) (-3, 2)

Ans. (3) (-3, 2)

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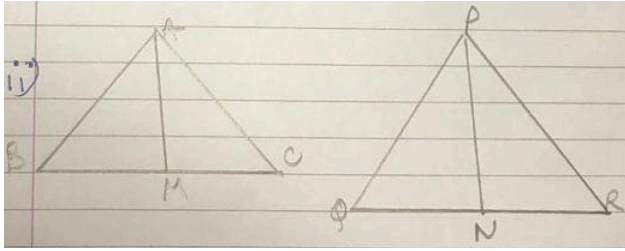
<sol_start id=38>

38. Area of kite = $2 \times (\text{area of } \triangle ABC)$

$$= 2 \left(\frac{1}{2} \times 6 \times 8 \right)$$

$$= 48 \text{ cm}^2$$

(ii) (1) RHS



(iii)

$$\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle PQR)} = \frac{AM^2}{PN^2}$$

$$\sqrt{\frac{4}{9}} = \frac{AM}{PN} \quad 2:3 = AM:PN$$

Ans. (1) 2:3

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